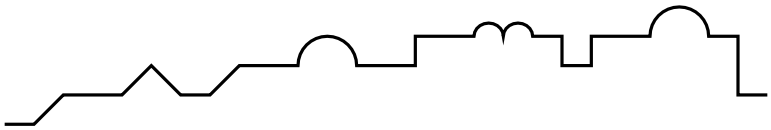
Math Kangaroo Practice

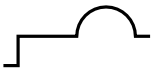
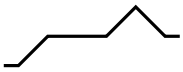
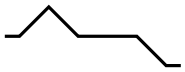
Grade 3-4 · 20 questions · harder · seed 23199

Part A: Each question is worth 3 points

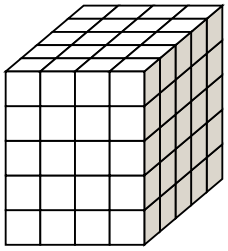
1. Far away we see the skyline of a castle.



Which of the pieces **cannot** belong to the skyline?

A.  ***B.***  ***C.*** ***D.*** ***E.*** 

2. Mia built this brick (4 cubes long, 5 cubes deep and 5 cubes tall) from red and blue cubes of the same size. The outside of the brick is completely red, but all the cubes inside are blue.

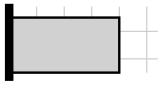


How many blue cubes did Mia use?

A. 82 ***B.*** 60 ***C.*** 48 ***D.*** 36 ***E.*** 18

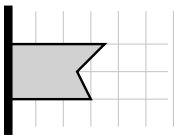
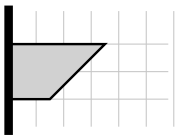
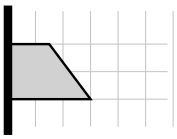
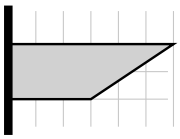
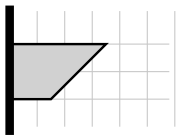
Part B: Each question is worth 4 points

3. From a window, Nina sees a rectangular flag on a pole. The wind turns and folds the flag, but the side sewn to the pole never moves. Nina draws the shadow of the flag on the wall at five different moments. When there is no wind at all, the flag hangs flat and is 4 squares wide and 2 squares tall (see the picture).



the flag, flat

Which shadow cannot be right?

A.  ***B.***  ***C.***  ***D.***  ***E.*** 

4. Lily has three cards with different digits. Two of them are 2 and 6; the third is face down. Using each card exactly once, with +, –, ×, ÷ and brackets, Lily gets 50.
Which of these digits can be on the hidden card?

A. 9 ***B.*** 8 ***C.*** 7 ***D.*** 5 ***E.*** 3

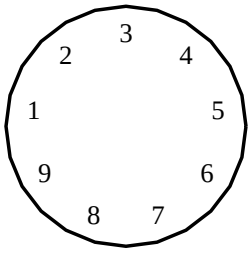
5. Leo builds a brick 4 cubes long and 3 cubes tall from red and blue cubes. The outside is completely red and the inside is completely blue. Leo used exactly 8 blue cubes.
How many cubes deep is the brick?

A. 2 ***B.*** 3 ***C.*** 4 ***D.*** 5 ***E.*** 6

6. Eva's snake toy is 12 dm 7 cm long. Ben's snake toy is 1120 mm long.
How many centimetres longer is Eva's snake?

A. 15 cm ***B.*** 22 cm ***C.*** 25 cm ***D.*** 116 cm ***E.*** 150 cm

7. A round cookie with the numbers 1 to 9 around its edge is cracked into 2 pieces. The sums of the numbers in the pieces are two consecutive numbers.



What is the largest of these sums?

A. 29 ***B.*** 25 ***C.*** 24 ***D.*** 23 ***E.*** 21

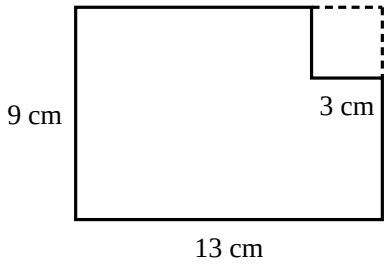
8. David has 2 T-shirts, 2 pairs of trousers and 3 pairs of shoes, all different. An outfit is one T-shirt and one pair of trousers and one pair of shoes.
How many different outfits can David make?

A. 4 ***B.*** 7 ***C.*** 8 ***D.*** 12 ***E.*** 13

9. Noah, Aisha, Zoe, Nicolas and Julien each play a different instrument: the violin, the xylophone, the trumpet, the guitar and the drum. Julien plays with sticks. Noah plays with sticks. Aisha plays an instrument with strings. Zoe uses a bow. Julien doesn't play the xylophone.
Which statement is true?

A. Nicolas plays the violin ***B.*** Zoe plays the xylophone ***C.*** Nicolas plays the trumpet
D. Aisha plays the drum ***E.*** Zoe plays the guitar

10. A rectangle is 13 cm long and 9 cm wide. A square with side 3 cm is cut off one corner.



What is the perimeter of the remaining shape?

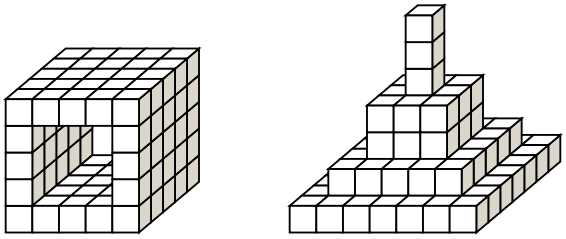
A. 56 cm ***B.*** 53 cm ***C.*** 50 cm ***D.*** 44 cm ***E.*** 41 cm

11. Five children, David, Clara, Anna, Iris and Ben, sit on a bench. David always sits at the left end and Clara always sits at the right end.
In how many different ways can the children sit on the bench?

A. 2 ***B.*** 3 ***C.*** 4 ***D.*** 5 ***E.*** 6

Part C: Each question is worth 5 points

12. Tom has the tunnel on the left. Tom takes it apart to build the pyramid on the right, but there are not enough cubes. Nothing is hollow inside the pyramid.



tunnel pyramid

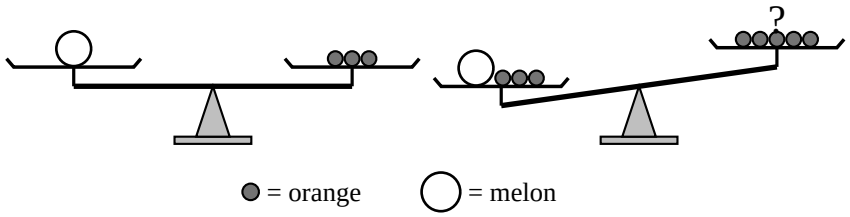
How many more cubes does Tom need?

A. 79 ***B.*** 15 ***C.*** 14 ***D.*** 12 ***E.*** 4

13. The odometer of a scooter shows 3479 km. Each digit is bigger than the digit before it.
After how many more kilometres will this happen again?

A. 10 ***B.*** 11 ***C.*** 20 ***D.*** 88 ***E.*** 89

14. The first balance is level. On the second balance there is 1 melon and 3 oranges on the left and 5 oranges on the right.



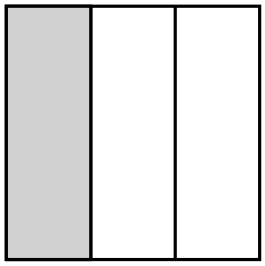
How many oranges must be added to the right plate to make it level?

A. 11 ***B.*** 6 ***C.*** 4 ***D.*** 3 ***E.*** 1

15. Tom, Lucas, Nora, Zoe and Fabienne each play a different instrument: the harp, the violin, the flute, the guitar and the drum. Tom doesn't play the guitar. Tom does not blow into the instrument. Zoe plays with sticks. Fabienne and Nora both play an instrument with strings. Fabienne uses a bow.
Which statement is **not** true?

A. Lucas plays the violin ***B.*** Nora plays the guitar ***C.*** Tom plays the harp
D. Fabienne plays the violin ***E.*** Zoe plays the drum

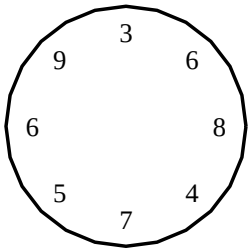
16. Mia cuts a square chocolate bar into 3 equal strips as shown. The perimeter of each strip is 48 cm.



What is the perimeter of the square?

A. 60 cm ***B.*** 72 cm ***C.*** 76 cm ***D.*** 96 cm ***E.*** 144 cm

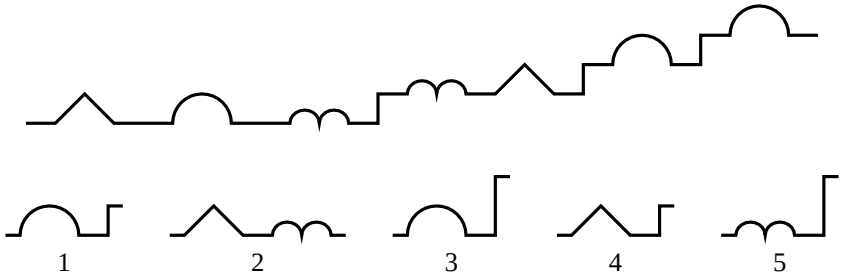
17. Numbers are written around a round plate as shown. Tom wants to break the plate with one straight line so that the two pieces have the same sum of numbers.



In how many different ways can he do it?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 4

18. This is the outline of a town seen from the river at sunset.



How many of the pieces 1–5 below belong to the line?

A. 0 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

19. The odometer of a scooter shows 2389 km. Each digit is bigger than the digit before it. After how many more kilometres will this happen again?

A. 67 ***B.*** 68 ***C.*** 69 ***D.*** 76 ***E.*** 77

20. Nina has the cards 1, 6, 8 and 9. Using every card exactly once, with +, –, ×, ÷ and brackets, Nina wants to get the biggest possible number.
What is the biggest number Nina can get?

A. 504 ***B.*** 531 ***C.*** 543 ***D.*** 564 ***E.*** 615

Answer Key

1.	**E**	**2.**	**E**	**3.**	**D**	**4.**	**B**	**5.**	**E**	**6.**	**A**
7.	**D**	**8.**	**D**	**9.**	**C**	**10.**	**D**	**11.**	**E**	**12.**	**B**
13.	**A**	**14.**	**E**	**15.**	**A**	**16.**	**B**	**17.**	**B**	**18.**	**B**
19.	**A**	**20.**	**A**								

Solutions

1. E — All the other pieces can be found in the line, but this one cannot: it is a mirror image of a part of the line. [2002 Q7 · cannot_belong]

2. E (18) — Take away one layer on every side: the inside is 2 × 3 × 3 = 18. [2003 Q9 · blue_inside]

3. D — Turning or folding can only make the flag look narrower. Picture D reaches more than 4 squares away from the pole, which is wider than the flag. [2002 Q15 · too_wide]

4. B (8) — 2 + 6 × 8 = 50. The other digits do not work with 2 and 6. [2002 Q21 · hidden_card]

5. E (6) — The blue inside is 2 long and 1 tall, so one layer of it has 2 × 1 = 2 cubes. 8 ÷ 2 = 4 layers of blue, plus the red front and back: 6. [2003 Q9 · how_deep]

6. A (15 cm) — 12 dm 7 cm = 127 cm and 1120 mm = 112 cm. 127 – 112 = 15 cm. [2003 Q10 · how_much_longer]

7. D (23) — All the numbers add up to 45. Two consecutive numbers that add up to 45: 22 + 23 = 45. The largest is 23. [2002 Q11 · largest_sum]

8. D (12) — Each T-shirt goes with each pair of trousers and each pair of shoes: 2 × 2 × 3 = 12. [2002 Q22 · outfits]

9. C (Nicolas plays the trumpet) — The only possible matching: Noah – xylophone; Aisha – guitar; Zoe – violin; Nicolas – trumpet; Julien – drum. [2002 Q18 · which_true]

10. D (44 cm) — The two new edges (3 cm and 3 cm) replace the two removed pieces of the outer edge, which were also 3 cm and 3 cm. So the perimeter is the same as the rectangle's: 2 × (13 + 9) = 44 cm. [2002 Q6 · corner_cut]

11. E (6) — Only the 3 children in the middle can change places: 3 × 2 × 1 = 6 ways. [2002 Q22 · ends_fixed]

12. B (15) — Tunnel: 16 × 5 = 80 cubes. Pyramid: 7 × 7 + 5 × 5 + 3 × 3 × 2 + 1 × 1 × 3 = 95. Missing: 95 – 80 = 15. [2002 Q20 · not_enough]

13. A (10) — Keep as many digits on the left as possible. The next such number is 3489: 3489 – 3479 = 10. [2002 Q24 · going_up]

14. E (1) — One melon = 3 oranges, so the left plate is worth 1 × 3 + 3 = 6 oranges. The right plate has 5, so add 6 – 5 = 1. [2002 Q9 · what_to_add]

15. A (Lucas plays the violin) — The only possible matching: Tom – harp; Lucas – flute; Nora – guitar; Zoe – drum; Fabienne – violin. [2002 Q18 · which_not_true]

16. B (72 cm) — A strip is the full side long and 1/3 of it wide, so its perimeter is side + side + 2 × (side ÷ 3) = 8 × (side ÷ 3). 48 ÷ 8 = 6, so the side is 3 × 6 = 18 cm and the perimeter is 72 cm. [2002 Q6 · strips_to_square]

17. B (1) — Each half must have 24. The ways: 8+4+7+5 | 6+9+3+6. [2002 Q11 · count_ways]

18. B (2) — Pieces 1, 4 fit. Not: 2 (it is a mirror image of a part of the line); 3 (its step is taller than in the line); 5 (its step is taller than in the line). [2002 Q7 · how_many_fit]

19. A (67) — Keep as many digits on the left as possible. The next such number is 2456: 2456 – 2389 = 67. [2002 Q24 · going_up]

20. A (504) — (1 + 6) × 8 × 9 = 504. Multiplying by 1 changes nothing, so it is better to add the 1 to the smallest other card first. [2002 Q21 · biggest]